

Chapter 4: Arrays in Java

Chapter 4 – Arrays in Java (3 Hours, Active Learning in Lab)

 **Session Duration: 3 hours**

 **Learning Style: Open-ended, Active Learning (Think-Pair-Share, Code-along, Mini Challenges)**

 **Lab-Based: Focus on writing and experimenting with code.**

Learning Objectives

By the end of this session, students should be able to:

1. Define and initialize one-dimensional and two-dimensional arrays.
 2. Pass arrays as arguments to methods.
 3. Return arrays from methods.
 4. Work with string arrays.
 5. Create and manipulate arrays of objects.
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Lecture & Lab (with code examples)

1. One-Dimensional Arrays

An array is a collection of variables of the same type, stored in a contiguous memory location. Let's create a simple array of integers.

Java Example:

Java

```
int[] numbers = {10, 20, 30, 40, 50};

for (int i = 0; i < numbers.length; i++) {
    System.out.println("Element at index " + i + ": " +
        numbers[i]);
}
```

Task 1:

Declares an array of **5 double values** representing the marks of a student in 5 subjects.

```
double [] marks = {86.5, 86.3, 87.8, 89.9, 92.3};
```

2. Passing Arrays to Methods

You can **pass an array** to a method like any object in Java. Let's pass our **numbers** array to a method that calculates the sum.

Java Example:

Unset

```
public static int sumArray(int[] arr) {
    int sum = 0;
    for (int num : arr) {
        sum += num;
    }
    return sum;
}
```

```
public static int findSmaller(int [] arr) {
    int min = arr[0];
    for (int num : arr) {
        if (num < min) {
            min = num;
        }
    }
    return min;
}
```

Task 2:

Create a method that returns the smallest value in an integer array.

3. Returning Arrays from Methods

Sometimes we want a method to return an entire array. For example, doubling each value in the array.

Java Example:

Unset

```
public static int[] doubleValues(int[] input) {  
    int[] result = new int[input.length];  
    for (int i = 0; i < input.length; i++) {  
        result[i] = input[i] * 2;  
    }  
    return result;  
}
```

```
public static int[] reverseArray(int[] input){  
    int[] reversed = new int[input.length];  
    for (int i=0; i<input.length; i++) {  
        reversed[i] = input[input.length-1-i];  
    }  
    return reversed;  
}
```

Challenge 1: Create a method that receives an array and returns a reversed version of that array.

4. String Arrays & Arrays of Objects

String Arrays : String arrays work the same way, but contain string objects.

Unset

```
String[] fruits = {"Apple", "Banana", "Cherry"};  
for (String fruit : fruits) {  
    System.out.println(fruit);  
}
```

Array of Objects Example:

Unset

```
class Student {  
    String name;  
    int age;  
    Student(String n, int a) {  
        name = n;  
        age = a;  
    }  
}
```

```
Student[] students = {
```

```
class Book {  
    String title;
```

```
    Book(String title) {  
        this.title = title;  
    }  
}
```



```

        new Student("Alice", 20),
        new Student("Bob", 22)
    };

    for (Student s : students) {
        System.out.println(s.name + " is " + s.age + " years old.");
    }
}

```

```

    }
    Book [] book = {
        new Book("Holly"),
        new Book("Loula")
    };
    for (Book b : book) {
        System.out.println(b.title);
    }
}

```

Challenge 2: Create a `Book` class. Create an array of 3 books and display their titles.

5. Two-Dimensional Arrays

A 2D array is like a grid (matrix). It's declared like `int[][] matrix`.

Unset

```

int[][] matrix = {
    {1, 2, 3},
    {4, 5, 6}
};

for (int i = 0; i < matrix.length; i++) {
    for (int j = 0; j < matrix[i].length; j++) {
        System.out.print(matrix[i][j] + " ");
    }
    System.out.println();
}

```

Challenge 3: Write a program that initializes a 3x2 matrix with random values and calculates the sum of each row.

Summary Notes

```

import java.util.Random;

public class matrixSum {
    public static void main (String[] args) {
        int [][] matrix = { {1,2,3}, {4,5,6} };

        for (int i=0; i<3; i++) {
            for (int j=0; j<2; j++) {
                matrix[i][j] = rand.nextInt(100);
            }
        }
    }
}

```

Java

```
// One-Dimensional Array  
int[] arr = new int[5]; // or {1,2,3}
```

```
// Passing array to method  
void printArray(int[] a) { ... }
```

```
// Returning array from method  
int[] createArray() { return new int[]{...}; }
```

```
// String array  
String[] names = {"Alice", "Bob"};
```

```
// Array of objects  
Student[] group = { new Student("A", 20), new Student("B", 21) };
```

```
// 2D array  
int[][] matrix = new int[3][3];
```

```
for (int i=0; i < matrix.length; i++) {  
    for (int j=0; j < matrix[i].length; j++) {  
        system.out.print(matrix[i][j] + " ");  
    }  
    system.out.println();  
}
```

✓ Optional (Challenge yourself)

- **Temperature Converter:** Array of temperatures, convert to Fahrenheit.
 - **Student Grades System:** Array of student objects with marks; calculate average.
 - **Tic-Tac-Toe Board:** Display using 2D array.
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Java

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// One-Dimensional Array
int[] arr = new int[5]; // or {1,2,3}

// Passing array to method
void printArray(int[] a) { ... }

// Returning array from method
int[] createArray() { return new int[]{...}; }

// String array
String[] names = {"Alice", "Bob"};

// Array of objects
Student[] group = { new Student("A", 20), new Student("B", 21) };

// 2D array
int[][] matrix = new int[3][3];
```

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